

Variable Conversion in C#

Chapter 3 · Part 3 · Type Casting & Conversion

Today we are going to learn about how to convert variables from one form to another. We know that there are four main variable types in C#: **int** (integer), **double** (decimal), **char** (single letter), and **string** (set of characters with a definite length). When we take user input, it is always in **string** form — we must convert it to the required type before using it in calculations.

Type	Full Name	Example	Convert Method
int	Integer	5, -3, 100	Convert.ToInt32()
double	Decimal	3.14, 20.5	Convert.ToDouble()
char	Character	'A', 'z'	Convert.ToChar()
string	Text	"Hello", "ABC"	Console.ReadLine()

♣ Integer to Decimal (int → double)

• Algorithm

1. Assign a value or take user input.
2. Convert it to **integer** using `Convert.ToInt32()`.
3. Convert the integer to **double** using `Convert.ToDouble()`.
4. Display the final result using `Console.WriteLine()`.

• Source Code — Static Method

```
// Static Method
using System;
class Program
{
    static void Main(string[] args)
    {
        int num = 20;
        int result = Convert.ToInt32(num);
        double res = Convert.ToDouble(result);
        Console.WriteLine(res); // Output: 20
    }
}
```

• Source Code — User Input Method

```
// User Input Method
using System;
class Program
{
    static void Main(string[] args)
    {
```

```
int num = Convert.ToInt32(Console.ReadLine());
double result = Convert.ToDouble(num);
Console.WriteLine(result); // Output: 20 (or .0 if shown)
}
}
```

♣ Decimal to Integer (double → int)

• Algorithm

1. Assign a value or take user input.
2. Store the decimal value as **double**.
3. Convert it to **integer** using `Convert.ToInt32()` — decimal part is truncated.
4. Display the final result using `Console.WriteLine()`.

• Source Code — Static Method (corrected)

```
// Static Method - Corrected
using System;
class Program
{
    static void Main(string[] args)
    {
        double num = 20.45;
        double result = Convert.ToDouble(num);
        int res = Convert.ToInt32(result);
        Console.WriteLine(res); // Output: 20 (decimal truncated)
    }
}
```

• Source Code — User Input Method

```
// User Input Method
using System;
class Program
{
    static void Main(string[] args)
    {
        double num = Convert.ToDouble(Console.ReadLine());
        int result = Convert.ToInt32(num);
        Console.WriteLine(result); // Output: 20 (if input was 20.45)
    }
}
```

♣ Important Notes

Type	Input Method	Description
string	<code>Console.ReadLine();</code>	Reads a full line of text
int	<code>Convert.ToInt32(Console.ReadLine());</code>	Reads and converts to integer
double	<code>Convert.ToDouble(Console.ReadLine());</code>	Reads and converts to decimal
char	<code>Console.Read();</code>	Reads a single character

■ **Common Mistake:** In C#, always use `string` (lowercase) not `String`. Also, `Convert.ToInt32(20.5)` truncates the decimal — result is 20, not 21.